hood. Thus the electroscope becomes charged, and its leaves spread apart. Of course, the apparatus in this rudimentary form is not sensitive enough to detect individual particles. Indeed a far more delicate apparatus, such as a string electrometer, while able to detect a rather weak radiation, cannot register individual particles. The method of detecting small charges has been improved considerably in various ways and respects, according to the particular nuclear-physical purpose in view.

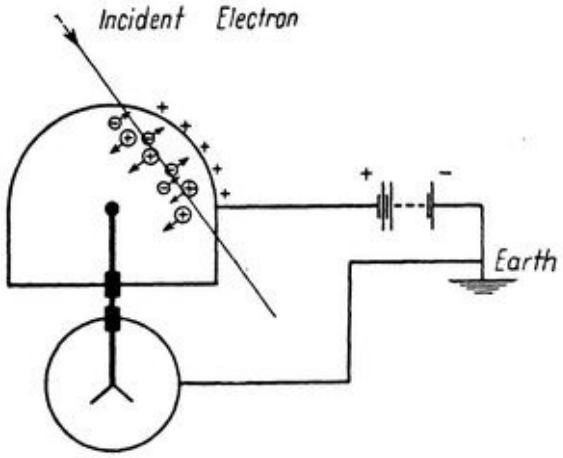


Figure 29.—The principle of the ionisation chamber.

The first apparatus of general applicability (for the detection of individual particles) was *Geiger's point counter* (Figure 30). Fundamentally it is simply a vastly improved ionization chamber. In this apparatus, a metal rod is drawn to a fine point, and the chamber is given a fairly high voltage, with the result that a strong electric field is created in the vicinity of the point. When a charged particle or a gamma-ray photon flies past and liberates electrons there, these liberated electrons are so strongly accelerated by the intense field that they, in turn, are able to tear electrons off air molecules. These electrons, in turn, are able to do the same thing, and thus the number of electrons liberated increases like an avalanche, and their number ceases to increase only where the field is weaker. But during this process, such enormous numbers of them are produced even by the effect of one single particle or photon that they can be detected by the means available to us.

